Tensor computations

Near-optimal type-2 bound for Gaussian sums of symmetric tensors

Difficulty: extreme **Importance:** broadly interesting

Statement

For a symmetric real tensor $T \in (\mathbb{R}^d)^{\otimes r}$, define

$$\|T\|_{\mathcal{I}_p} = \max_{\|x\|_p \le 1} |\langle T, x^{\otimes r}\rangle|.$$

For every integer $r \ge 2$ and real $2 \le p < \infty$, do constants $C_{r,p} > 0$ and $a_{r,p} \ge 0$ exist such that, for every $d \ge 2$, $N \ge 1$, and deterministic symmetric tensors $T_1, \dots, T_N \in (\mathbb{R}^d)^{\otimes r}$,

$$\mathbb{E}\left\|\sum_{i=1}^N g_i T_i\right\|_{\mathcal{I}_p} \le C_{r,p}\, d^{1/2-1/p} [\log(2+dN)]^{a_{r,p}} \left(\sum_{i=1}^N \|T_i\|_{\mathcal{I}_p}^2\right)^{1/2},$$

where g_i are independent real $N(0,1)$ random variables? Constants and logarithmic exponents must be independent of d, N and the tensors. The explicit logarithmic form spells out the source's $\widetilde{O}_{r,p}$ notation.

Relevance

This would control the injective norm of tensor perturbations with general covariance, extending a central matrix concentration principle to multilinear computation and noisy tensor models.

References

1. A. S. Bandeira, D. Dmitriev, K. Lucca, P. Nizić-Nikolac, and A. Rödder, *Randomstrasse101: Open Problems of 2025*, arXiv:2603.29571, 2026. Primary text, Entry 8 by K. Lucca, Conjecture 16 and equations (1)–(2).
2. A. S. Bandeira, S. Gopi, H. Jiang, K. Lucca, and T. Rothvoss, *A Geometric Perspective on the Injective Norm of Sums of Random Tensors*. Primary preprint, introduction and tensor type-2 bounds; published as *Tensor Concentration Inequalities: A Geometric Approach*, STOC 2025.

Status check — 2026-09-08

Reference 1's displayed manuscript is dated August 24, 2026 and retains the conjecture. It states that $p \ge 2r$ is proved, as is the matrix case $r = p = 2$; general $p < 2r$ remains beyond the available bounds. Searches *"tensor" "type-2" "conjecture" "2026"* and *"tensor" "type-2" conjecture proof 2025 2026* found no full resolution. Results for independent entries or rank-one summands impose stronger hypotheses. The $p = \infty$ endpoint is already covered by the source's established high-p regime and is not an additional open entry.